

**Exploring Excess Return Generation
through European Fixed Income Arbitrages**

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Abstract

We construct systematic trading strategies on three European fixed-income arbitrages that the literature has documented as pricing anomalies but never traded: the BTP Italia inflation-linked basis, the corporate CDS–bond basis, and the iTraxx index skew. Net of regime-dependent transaction costs, the strategies earn annualized Sharpe ratios of 0.98 to 1.56 with positive skewness and drawdowns below 4.5%. The alpha—2.2% to 4.8% per year—survives three canonical factor benchmarks, principal components, and best-subset selection from 75 tradable factors, in and out of sample; model-X knockoffs at a 10% false discovery rate retain no factor at all. The premium doubles to triples when intermediary funding stress is high, and a single funding shock moves the two institutionally held mispricings wider while moving the retail-held BTP Italia basis tighter—direct evidence of slow-moving capital and investor-base segmentation. For investors, the three trades combine a state-dependent premium with an internal hedge, and post-crisis regulation makes the mispricings structural rather than transient.

1 Introduction

Duarte et al. (2007) ask whether fixed-income arbitrage earns genuine alpha or merely picks up “nickels in front of a steamroller.” We take their question to European markets and to three strategies that have each been documented as a pricing anomaly but never examined as a systematic trading strategy: the inflation-linked basis on Italian BTP Italia bonds, the European corporate CDS–bond basis, and the iTraxx index skew—the gap between the traded index spread and the spread implied by its single-name constituents.

Three features make these trades unusually clean laboratory material. First, each has a deterministic payoff at maturity: held to maturity, the profit is locked in regardless of the interim path—unlike the model- and assumption-dependent strategies of Duarte et al. (2007). Second, all three are implementable: entry and exit follow mechanical basis thresholds, transaction costs are charged at the trade level and widen with market stress, and every reported return is net of those costs. Third, and central to this paper, the three trades are run by the same type of leveraged institutional arbitrageur but differ in who *holds* the underlying instruments: the CDS–bond and iTraxx legs sit on institutional balance sheets, whereas the BTP Italia is, by design of the Italian Treasury, a retail instrument held by unlevered households. This heterogeneity turns the cross-section into a natural experiment on the slow-moving capital theory of Duffie (2010): if what closes mispricings is a common pool of intermediary capital (Brunnermeier and Pedersen, 2009; Mitchell and Pulvino, 2012), then trades sharing that pool should co-move, the co-movement should intensify in stress, and the trade whose holders face no margin calls should behave differently—identification that a common macroeconomic factor cannot mimic.

Our findings, in brief. Net of costs, the strategies earn annualized Sharpe ratios of 0.98 (BTP Italia), 1.56 (CDS–bond), and 1.25 (iTraxx skew), with positive skewness—the opposite of the steamroller caricature—and maximum drawdowns of 3.0%, 4.4%, and 2.5%, each recovered within months. The alpha, 2.2% to 4.8% per year, survives every layer of an escalating battery of risk controls: the benchmark models of Duarte et al. (2007), Fung and Hsieh (2004), and Brooks et al. (2020); eight principal components of a 75-factor tradable universe; and best-subset selection that is free to pick, for each strategy, whichever eight factors explain it best. The selected models absorb up to 31% of return variation and none of the alpha, which also survives when every hedge is re-estimated out of sample from past-only loadings. The alpha is state-dependent: it rises two- to sevenfold from calm to stressed regimes. And the slow-moving capital predictions hold with the sign structure the holder-base experiment demands: the institutional pair co-moves ever more tightly as stress builds (correlation rising from -0.21 to $+0.32$), while the retail-held BTP Italia moves *against* the institutional pair, reaching -0.47 in high stress; a widening of the Libor–OIS spread pushes the two institutional mispricings wider and the BTP Italia basis tighter.

The remainder of the paper follows this arc: Section 2 constructs the strategies and measures performance; Section 3 runs the alpha through five layers of controls; Section 4 tests the slow-moving capital mechanism; Section 5 draws the implications for investors. Construction details, the full factor universe, and all robustness exhibits are in the Appendix.

Table 1. Performance of the three arbitrage strategies, net of transaction costs. Monthly signal-weighted excess returns; mean and volatility annualized; t -statistic Newey–West (4 lags); MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007) at $\rho = 3$. Samples end May 2025.

Strategy	n	Mean (% p.a.)	t	Vol. (% p.a.)	Sharpe	Skew	Max DD (%)	MPPM (% p.a.)
BTP Italia	163	1.59	3.52	1.62	0.98	0.81	−3.0	1.48
CDS–Bond Basis	207	4.61	5.87	2.96	1.56	0.79	−4.4	4.39
iTraxx Skew	243	1.91	4.92	1.53	1.25	1.01	−2.5	1.87

2 Three European Fixed-Income Arbitrages

Each arbitrage is defined by a basis that no-arbitrage forces toward a well-defined level, and each trade is a fictional fund, in the design of Duarte et al. (2007), that opens when the basis crosses an entry threshold and closes on reversion or maturity.

BTP Italia. The BTP Italia is indexed to Italian inflation and differs from conventional linkers in two ways that matter here. It pays the inflation revaluation with each semiannual coupon and redeems at par, so the notional at risk never balloons and the two legs of the basis trade carry identical sovereign credit exposure; and it is placed primarily to retail households, with a dedicated retail-only auction phase and a loyalty bonus for buy-and-hold investors. The basis is the fixed-rate-equivalent yield of the BTP Italia (converted with zero-coupon inflation swaps, following the replication logic of Fleckenstein et al., 2014) minus the yield of the maturity-matched nominal BTP; both legs fund at the same general-collateral repo rate.

CDS–bond basis. For each issuer in the on-the-run iTraxx Main universe, the basis is the CDS spread minus the bond z-spread. When negative, buying the bond and buying protection locks in the gap and returns par in default, since the CDS tenor matches the bond’s residual maturity; we trade the negative basis only, because the reverse trade requires a corporate reverse repo subject to recall risk in stress (Bai and Collin-Dufresne, 2019).

iTraxx skew. The skew is the traded index spread minus the intrinsic spread bootstrapped from the constituents’ CDS curves; we trade it across the four iTraxx sub-indices, on-the-run only, both legs under a single netting set. It is a derivative-on-derivative trade with no cash leg to finance.

Individual trades aggregate into signal-weighted indices, in which weight is proportional to the entry dislocation from the historical equilibrium (equal-weighted results, which are very close, are in Appendix A). Transaction costs are charged per trade in proportion to DV01, at fee tiers that rise with the level of the iTraxx Main 5-year spread—LOW below 60 bps, MEDIUM to 100 bps, HIGH above—so costs are most conservative exactly when arbitrage returns are largest. All results are net of these costs.

Table 1 reports the record. The samples run from April 2012 (BTP Italia), September 2008 (CDS–bond), and September 2005 (iTraxx skew) through May 2025, generating 260, 931, and 206 trades. The three Sharpe ratios sit between 0.98 and 1.56 with positive skewness throughout, and the manipulation-proof measure of Goetzmann et al. (2007)—reported at risk aversion $\rho = 3$ and nearly invariant to ρ —confirms that the risk-adjusted performance is not manufactured by the dynamic trading rule or by tail risk. Drawdowns are shallow and short: the deepest, 4.4% for the CDS–bond basis in 2008–09, is an order of magnitude smaller than the strategy’s cumulative return over the sample.

3 Is It Alpha? Five Layers of Risk Controls

The question is whether the returns of Table 1 are compensation for systematic risk. We answer it with an escalating battery: if the alpha is an artifact of an omitted exposure, some layer should absorb it.

The first three layers are the canonical benchmarks of the fixed-income arbitrage and active-management literatures—the ten-factor model of Duarte et al. (2007), the seven-factor model of Fung and Hsieh (2004), and the eight risk premia of Brooks et al. (2020)—estimated on euro counterparts of every factor (U.S.-factor versions, with the same conclusions, in Appendix B). The fourth layer replaces specified factors with the first eight principal components of a candidate universe of 75 tradable risk factors spanning credit, liquidity,

Table 2. Annualized alpha (% p.a.) across five layers of factor controls. Panel A: full-sample spanning regressions (Newey–West t -statistics; adjusted R^2 in brackets). Panel B: out-of-sample alpha with hedge loadings estimated on past-only expanding windows (60-month burn-in). ***, **, *: 1%, 5%, 10%. All best-subset out-of-sample t -statistics exceed the Harvey et al. (2016) threshold of 3.0.

Control layer	BTP Italia	CDS–Bond	iTraxx Skew
<i>Panel A: In-sample alpha [adj. R^2]</i>			
Duarte et al. (2007), $k=10$	+1.6*** [0.11]	+4.2*** [0.15]	+2.2*** [0.06]
Brooks et al. (2020), $k=8$	+2.1*** [0.08]	+4.2*** [0.12]	+1.7*** [0.04]
Fung–Hsieh (2004), $k=7$	+1.8*** [0.10]	+4.0*** [0.11]	+1.8*** [0.00]
Principal components, $K=8$	+1.4*** [0.09]	+4.8*** [0.11]	+2.2*** [0.13]
Best subset (ℓ_0), $k=8$	+2.2*** [0.31]	+4.8*** [0.21]	+2.9*** [0.25]
<i>Panel B: Out-of-sample alpha (expanding window)</i>			
Duarte et al. (2007)	+1.29**	+3.48***	+2.27***
Brooks et al. (2020)	+2.10***	+3.86***	+1.77***
Fung–Hsieh (2004)	+1.69***	+3.45***	+2.02***
Principal components	+1.61***	+4.55***	+1.85***
Best subset (frozen set)	+2.05***	+3.90***	+2.50***

equity, volatility, and rates. The fifth is best-subset selection (Bertsimas et al., 2016): for each strategy, an ℓ_0 -constrained search picks whichever eight of the candidates explain its returns best—the hardest test we can pose, because the data are free to construct the most absorbing hedge.

Table 2 reports the outcome. Moving down the rows, explanatory power rises from near zero to adjusted R^2 of 21–31%, while the alpha barely moves: 1.4–2.2% per year for BTP Italia, 4.0–4.8% for the CDS–bond basis, 1.7–2.9% for the iTraxx skew, significant at the 1% level in every cell. Panel B repeats the exercise in real time: at each month the hedge is re-estimated on a past-only expanding window (Welch and Goyal, 2008), so the surviving alpha is an abnormal return an investor could actually have earned. It survives everywhere, and for the best-subset hedge every out-of-sample t -statistic clears the 3.0 hurdle that Harvey et al. (2016) propose for discoveries in multiple-testing environments.

Three details sharpen the verdict. First, the three selected factor sets are pairwise disjoint—no factor appears in more than one strategy—so no single common exposure is responsible; the economically telling loading is the iTraxx skew’s on the traded intermediary-capital factor of He et al. (2017): the strategy pays off when dealer equity underperforms. Second, the result is invariant to the selector: five alternatives, from the adaptive elastic net (Zou and Zhang, 2009) to a random-forest screen, leave the alpha intact under honest split-sample inference, and the most conservative of all—model-X knockoffs (Candès et al., 2018) at a 10% false discovery rate—select *no factor at all*: the data neither identify a stable factor structure nor produce one that absorbs the premium (Appendix C). Third, the alpha is state-dependent. In the conditional framework of Christopherson et al. (1998), a one-standard-deviation rise in the (lagged) iTraxx Main spread adds 1.7–3.2 percentage points of annual alpha to the CDS–bond basis and 0.9–1.5 to the iTraxx skew; split discretely by regime, the high-stress alpha is two to seven times the calm-market alpha for every strategy and every control set (Appendix B).

An alpha that no factor spans, that survives in real time, and that concentrates in funding stress is not a risk premium in disguise; it is the fingerprint of an arbitrage that capital fails to close precisely when closing it is most profitable. The next section asks why.

4 Slow-Moving Capital and the Holder-Base Experiment

Why does an alpha of 2–5% per year survive in liquid, investment-grade European markets? The slow-moving capital theory (Duffie, 2010; Brunnermeier and Pedersen, 2009; Mitchell and Pulvino, 2012) gives a falsifiable answer: mispricings persist because the intermediary capital that closes them is scarce, shared, and slow to redeploy. Our three-strategy setting embeds an unusually sharp test. All three trades are run by the same type of leveraged arbitrageur, but the CDS–bond and iTraxx legs are *held* by institutions, while the BTP Italia bond is held by unlevered retail households who face no margin calls and do not sell when

dealer capital withdraws (Vayanos and Vila, 2021). If a common capital pool is the mechanism, the two institutional mispricings should co-move and tighten their link in stress, while the retail-held basis should sit outside the pool—and if anything move the other way, as forced institutional selling in the two credit trades coincides with a flight of household demand toward the linker. A common macroeconomic shock cannot generate that sign structure: it would move all three claims alike, whoever holds them.

Table 3, Panel A, shows the sign structure in the co-movement of monthly mispricing changes. The institutional pair correlates at +0.19 unconditionally, and the correlation rises monotonically with stress, from −0.21 in the LOW regime to +0.32 in HIGH—an increase significant at the 1% level (Fisher z), and robust at the 10% level to the Forbes and Rigobon (2002) correction for stress-driven volatility, so it is a genuine tightening of the cross-market link rather than a variance artifact. The BTP Italia pairs behave as the experiment predicts: the correlation with the CDS–bond basis is −0.30 on the full sample and deepens to −0.47 in high stress, and an interaction regression confirms the asymmetry—stress *amplifies* transmission within the institutional pair ($\beta_1 = +0.08$, 5% level) and *deepens the negative* link toward the retail-held basis ($\beta_1 = -0.64$, 1% level). Dynamic conditional correlations tell the same story month by month: the BTP–CDS pair is negative in every single month of the sample (Appendix D).

Timing and persistence complete the mechanism. Granger tests on a VAR of the three mispricings recover the liquidity pecking order of Mitchell and Pulvino (2012): shocks originate in the most liquid, standardized market—the iTraxx index, where dealer inventory adjusts fastest—and propagate to the cash CDS–bond basis ($p = 0.011$) and the BTP Italia basis ($p = 0.007$) within a month, never the reverse; generalized impulse responses trace the same liquid-to-illiquid path with the same signs, positive into the CDS–bond basis and negative into BTP Italia (Appendix D). Persistence lines up with funding dependence, the one margin on which the three trades differ mechanically: the CDS–bond basis is the only trade that must finance a cash bond on a dealer balance sheet for the life of the position, and its mispricing half-life, 3.0 months, is one-third longer than the 2.2 months of the BTP Italia and the iTraxx skew—which need no such financing and are, on funding grounds, indistinguishable in persistence.

Panel B closes the loop by regressing each mispricing on observable proxies for the shadow cost of intermediary funding. The Libor–OIS spread is significant in all three equations and *reverses sign across holder bases*: +16.63 on the CDS–bond basis and +3.85 on the iTraxx skew, both at the 1% level, against −15.99 on BTP Italia; market illiquidity repeats the reversal. Dealer credit risk loads only on the one trade that must be financed outright. A margin constraint binds only where the holder is levered—an aggregate risk premium would move two claims on the same cash flows in the same direction whoever holds them—so the sign reversal is the signature of the funding channel itself.

Why, then, is the premium not competed away? The limits differ by trade but share one root: the post-crisis cost of intermediary balance sheet. The CDS–bond trade consumes repo capacity in a thin corporate-repo market subject to recall risk (Bai and Collin-Dufresne, 2019; Mitchell and Pulvino, 2012); the iTraxx skew consumes regulatory capital—Boyarchenko et al. (2016) document that as required leverage ratios rose from 2–3% to 5–6%, the return on equity of index-versus-constituent trades fell by a factor of two to three, producing “new normal” basis levels; and the BTP Italia basis rests on a retail holder base that neither monitors nor arbitrages it. The alpha of Section 3 is not an anomaly awaiting correction; it is the equilibrium rent on a permanently scarcer balance sheet.

5 Implications for Investors

[ALEX: paragraph 1 – state-dependent deployment.]

[ALEX: paragraph 2 – BTP Italia as the intra-book hedge.]

[ALEX: paragraph 3 – structural persistence and monitoring.]

[ALEX: paragraph 4 – implementation and risk management.]

6 Conclusion

Three European fixed-income arbitrages—each documented as a pricing anomaly, none previously traded systematically—earn Sharpe ratios of 0.98 to 1.56 net of costs, with alpha of 2.2% to 4.8% per year that no layer of factor controls absorbs: not three canonical benchmarks, not eight principal components, not

Table 3. The holder-base experiment. Panel A: Pearson correlations of monthly mispricing changes Δm , unconditional and by stress regime (iTraxx Main 5Y: LOW < 60, MEDIUM 60–100, HIGH ≥ 100 bps); the LOW-to-HIGH rise for the institutional pair is significant at 1% (Fisher z , $p = 0.005$) and at 10% after the Forbes–Rigobon correction ($p = 0.070$). Panel B: each Δm regressed on proxies for the shadow cost of intermediary funding, on the 148 common months; Newey–West t -statistics. ***, **, *: 1%, 5%, 10%.

<i>Panel A: Co-movement of mispricing changes</i>				
Pair	Uncond.	LOW	MEDIUM	HIGH
CDS–Bond vs. iTraxx (institutional pair)	+0.19*	−0.21*	+0.20*	+0.32**
CDS–Bond vs. BTP Italia (across bases)	−0.30***	−0.13	−0.32***	−0.47**
BTP Italia vs. iTraxx (across bases)	−0.10	+0.15	−0.15	−0.14
<i>Panel B: Response to intermediary funding conditions</i>				
Driver	CDS–Bond	iTraxx Skew	BTP Italia	
Libor–OIS spread	+16.63***	+3.85***	−15.99**	
Amihud illiquidity	+6.93**	−0.42	−20.51***	
Dealer CDS spread ($\Delta 5Y$)	+0.11*	+0.02	−0.06	
HKM capital ratio	+6.50	−0.67	+8.81	
Adj. R^2	0.165	0.058	0.126	

a best-subset search over 75 tradable candidates, in sample or out. The premium concentrates in funding stress, and its cross-sectional behavior carries the signature of slow-moving capital: the two institutionally held mispricings co-move ever more tightly as intermediary capital tightens, while the retail-held BTP Italia basis moves against them—a sign structure that a common risk factor cannot produce and that identifies the balance-sheet channel of Duffie (2010). For investors, the package is a state-dependent premium with an internal hedge, kept open by post-crisis regulation rather than by risk. The geography of arbitrage capital, not the asset-class label, determines what co-moves with what—and, for those able to supply balance sheet when it is scarce, what still pays.

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A Strategy Construction and Additional Performance

Entry and exit thresholds, trade-level statistics, and the regime-dependent transaction-fee schedule are reported below, followed by the equal-weighted counterparts of the signal-weighted indices used in the text and the volatility-managed diagnostics of Moreira and Muir (2017).

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Table A.1. Manipulation-Proof Performance Measure by Risk Aversion

MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007), computed from monthly returns; ρ is the coefficient of relative risk aversion and the risk-free rate is 1-month Euribor. Returns are net of transaction costs.

Strategy	$\rho = 2$ (% p.a.)	$\rho = 3$ (% p.a.)	$\rho = 4$ (% p.a.)
BTP Italia	1.50	1.48	1.47
CDS–Bond Basis	4.43	4.39	4.35
iTraxx Skew	1.88	1.87	1.86

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B Benchmark Factor Models

Full euro-panel estimates of the three benchmark specifications, the consolidated alpha synthesis (unconditional, conditional, out-of-sample), and the alpha-by-regime decompositions.

Table A.2. Duarte Factor Model Regressions

This table reports estimates of the regression $r_{i,t} = \alpha_i + \beta_1 Mkt_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 R_{S,t} + \beta_6 R_{I,t} + \beta_7 R_{B,t} + \beta_8 R_{2,t} + \beta_9 R_{5,t} + \beta_{10} R_{10,t} + \varepsilon_{i,t}$, where $r_{i,t}$ is the monthly excess return of strategy i and the regressors are the ten Duarte et al. (2007) euro-area equity and bond risk factors. α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). Mkt : European value-weighted market excess return over the 1-month German bill; SMB , HML , UMD : the Fama–French European size, value, and momentum factors. R_S : excess return on the EuroStoxx Banks index. R_I , R_B : excess returns on European A-rated industrial and European bank bond indices. R_2 , R_5 , R_{10} : excess returns on European 2-, 5-, and 10-year government bond portfolios. $\bar{R}^2$ is the adjusted R^2 . Sample: BTP Italia 163 months, CDS–Bond 207 months, iTraxx 243 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	α (% p.a.)	β_{Mkt}	β_{SMB}	β_{HML}	β_{UMD}	β_{R_S}	β_{R_I}	β_{R_B}	β_{R_2}	β_{R_5}	$\beta_{R_{10}}$	$\bar{R}^2$
<i>BTP Italia</i>	1.62*** (3.29)	−0.04* (−1.79)	−0.06*** (−2.86)	−0.03 (−1.42)	0.02 (1.13)	0.01 (1.01)	−0.03 (−0.27)	0.04 (0.49)	0.03 (0.06)	−0.04 (−0.14)	−0.02 (−0.16)	0.11
<i>CDS–Bond</i>	4.24*** (5.27)	0.00 (0.20)	0.04 (0.90)	0.03 (1.11)	−0.02 (−0.88)	−0.02* (−1.74)	0.45*** (3.94)	−0.06 (−1.57)	−0.27 (−0.45)	−0.04 (−0.10)	−0.10 (−0.71)	0.15
<i>iTraxx</i>	2.19*** (4.90)	−0.02 (−1.25)	−0.03** (−2.04)	−0.07** (−2.26)	−0.02* (−1.95)	0.01 (1.53)	0.05 (0.75)	−0.02 (−0.78)	0.06 (0.16)	0.05 (0.21)	−0.06 (−0.78)	0.06
<i>Skew</i>												

Table A.3. Fung & Hsieh Factor Model Regressions

This table reports estimates of the regression $r_{i,t} = \alpha_i + \beta_1 S\&P_t + \beta_2 SC-LC_t + \beta_3 BdOpt_t + \beta_4 FXOpt_t + \beta_5 ComOpt_t + \beta_6 10Y_t + \beta_7 CredSpr_t + \varepsilon_{i,t}$, where $r_{i,t}$ is the monthly excess return of strategy i and the regressors are the seven Fung & Hsieh (2004) euro-area hedge fund risk factors. α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). S&P: European value-weighted market excess return. SC–LC: the Fama–French European size (small-minus-big) factor. BdOpt, FXOpt, ComOpt: trend-following returns from lookback straddles on government-bond, currency, and commodity futures. 10Y: excess return on a 10-year German government bond total-return index. CredSpr: return spread between euro-area BBB corporate bonds and 10-year German Bunds. $\bar{R}^2$ is the adjusted R^2 . Sample: BTP Italia 163 months, CDS–Bond 207 months, iTraxx 243 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	α (% p.a.)	$\beta_{S\&P}$	β_{SC-LC}	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	$\beta_{CredSpr}$	$\bar{R}^2$
<i>BTP Italia</i>	1.81*** (4.08)	−0.03** (−2.53)	−0.06*** (−2.70)	0.00 (1.25)	0.00 (0.03)	0.00 (1.09)	−0.01 (−0.27)	0.02 (0.30)	0.10
<i>CDS–Bond</i>	3.98*** (5.05)	−0.02 (−1.18)	0.05 (1.24)	0.00 (1.01)	−0.01* (−1.95)	−0.00 (−0.42)	0.17*** (3.43)	0.19*** (2.87)	0.11
<i>iTraxx</i>	1.84*** (4.49)	0.00 (0.41)	−0.02 (−1.50)	0.00 (0.60)	0.00 (1.01)	0.00 (0.57)	0.03 (0.83)	0.02 (0.62)	0.00
<i>Skew</i>									

Table A.4. Active Fixed Income Factor Model Regressions

This table reports estimates of the regression $r_{i,t} = \alpha_i + \beta_1 \text{Term}_t + \beta_2 \text{Global Term}_t + \beta_3 \text{Global Agg}_t + \beta_4 \text{Infl. Linkers}_t + \beta_5 \text{Corp Credit}_t + \beta_6 \text{Emerg Debt}_t + \beta_7 \text{Emerg Curr}_t + \beta_8 \text{Rates Vol}_t + \varepsilon_{i,t}$, where $r_{i,t}$ is the monthly excess return of strategy i and the regressors are the eight Active Fixed Income risk premia of Brooks et al. (2020). α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). Term: euro-area government bond excess return over cash. Global Term: Bloomberg Global Treasury Index (hedged), in excess of cash. Global Agg: Bloomberg Global Aggregate Bond Index (hedged), in excess of cash. Infl. Linkers: global inflation-linked bonds (hedged), in excess of cash. Corp Credit: pan-European high-yield corporate bond index, in excess of cash. Emerg Debt: excess return on hard-currency emerging-market debt (hedged). Emerg Curr: excess return on a basket of emerging-market currencies against the U.S. dollar. Rates Vol: excess return on a one-month variance swap on the ten-year Bund future, in percentage points of annualised variance. $\bar{R}^2$ is the adjusted R^2 . Sample: BTP Italia 163 months, CDS–Bond 207 months, iTraxx 243 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	α (% p.a.)	β_{Term}	β_{GlobTerm}	β_{GlobAgg}	$\beta_{\text{InflLinkers}}$	$\beta_{\text{CorpCredit}}$	$\beta_{\text{EmergDebt}}$	$\beta_{\text{EmergCurr}}$	$\beta_{\text{Rates Vol}}$	$\bar{R}^2$
<i>BTP Italia</i>	2.12*** (3.84)	-0.01 (-0.47)	-0.05 (-0.16)	0.13 (0.49)	-0.06 (-1.60)	-0.12** (-2.23)	0.05 (0.97)	-0.01 (-0.30)	-0.16 (-1.36)	0.08
<i>CDS–Bond</i>	4.20*** (5.20)	0.08* (1.89)	-0.43 (-0.73)	0.30 (0.56)	-0.07 (-1.05)	-0.03 (-0.53)	0.14*** (3.15)	0.01 (0.16)	0.32 (1.15)	0.12
<i>iTraxx Skew</i>	1.75*** (4.24)	-0.04* (-1.92)	-0.09 (-0.42)	0.33 (1.42)	-0.00 (-0.01)	0.02 (1.61)	-0.08** (-2.18)	0.01 (0.55)	0.19 (1.53)	0.04

Table A.5. Alpha Estimates Across Benchmark Factor Models

Panel A reports annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors (4 lags) using EUR factors. Panel B reports the intercept α_0 and the stress-loading coefficient α_1 from the Christopherson et al. (1998) full conditional model $r_t = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' X_t + \varepsilon_t$, in which both the alpha and the factor loadings are conditioned on z_{t-1} , the one-month-lagged standardized iTraxx Main 5Y spread (a predetermined conditioning variable); $\bar{R}^2$ is the adjusted R^2 . Significance of α_1 is assessed by a moving-block bootstrap ($B = 9,999$, $H_0: \alpha_1 = 0$ imposed, block length 9). Panel C reports out-of-sample alpha in the recursive design of Welch and Goyal (2008): loadings estimated by OLS on the expanding past-only window $[1, t]$ (60-month burn-in), and the realized abnormal return $e_{i,t} = r_{i,t} - \hat{\beta}'_{i,t-1} \mathbf{f}_t$ tested with Newey–West HAC; IR is the annualized information ratio of the hedged return, and N the number of out-of-sample months, common to the three benchmarks within each strategy (rolling-window counterpart: Appendix ??). t -statistics in parentheses are Newey–West HAC throughout. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel A: Full-sample alpha (% p.a.)

Strategy	Duarte et al. (2007)			Brooks et al. (2020)			Fung & Hsieh (2004)		
	α	t	N	α	t	N	α	t	N
<i>BTP Italia</i>	+1.62***	3.29	163	+2.12***	3.84	163	+1.81***	4.08	163
<i>iTraxx Skew</i>	+2.19***	4.90	243	+1.75***	4.24	243	+1.84***	4.49	243
<i>CDS–Bond Basis</i>	+4.24***	5.27	207	+4.20***	5.20	207	+3.98***	5.05	207

Panel B: Christopherson, Ferson, and Glassman (1998) conditional alpha (% p.a.)

	Duarte et al. (2007)			Brooks et al. (2020)			Fung & Hsieh (2004)		
	α_0	α_1	$\bar{R}^2$	α_0	α_1	$\bar{R}^2$	α_0	α_1	$\bar{R}^2$
<i>BTP Italia</i>	+1.58*** (3.78)	+1.20** (2.25)	0.23	+1.91*** (3.03)	+1.02 (1.14)	0.13	+1.70*** (5.24)	+0.50 (1.22)	0.19
<i>iTraxx Skew</i>	+1.94*** (4.75)	+0.93** (2.55)	0.11	+1.82*** (4.43)	+1.43*** (3.40)	0.14	+1.89*** (4.85)	+1.51*** (4.66)	0.05
<i>CDS–Bond Basis</i>	+4.67*** (5.99)	+2.10** (2.74)	0.25	+4.27*** (6.24)	+3.18*** (4.02)	0.18	+3.96*** (5.45)	+1.73** (2.47)	0.17

Panel C: Out-of-sample alpha (% p.a.)

	N	Duarte et al. (2007)			Brooks et al. (2020)			Fung & Hsieh (2004)		
		α	t	IR	α	t	IR	α	t	IR
<i>BTP Italia</i>	103	+1.29**	2.24	+0.78	+2.10***	3.83	+1.25	+1.69***	2.86	+1.05
<i>iTraxx Skew</i>	183	+2.27***	4.71	+1.37	+1.77***	3.48	+1.07	+2.02***	4.05	+1.20
<i>CDS–Bond Basis</i>	147	+3.48***	4.29	+1.25	+3.86***	4.30	+1.31	+3.45***	3.87	+1.19

C Factor Universe and Selection Robustness

The 75-factor candidate universe with sources, the PCA spanning regressions, the post-selection OLS on the best-subset factors, and the alpha across the five alternative selection methods with honest split-sample inference.

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Table A.6. PCA Spanning Regressions

This table reports estimates of the spanning regression $r_{i,t} = \alpha_i + \beta_1 PC_{1,t} + \beta_2 PC_{2,t} + \dots + \beta_8 PC_{8,t} + \varepsilon_{i,t}$, where $r_{i,t}$ is the monthly excess return of strategy i and the regressors are the first 8 principal components of the standardized candidate factor panel, extracted over the full estimation sample (the number of components is selected by the Bai–Ng IC_{p2} criterion). α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC, $[4(T/100)^{2/9}] = 4$ lags). $\bar{R}^2$ is the adjusted R^2 ; N is the number of monthly observations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	α (% p.a.)	β_{PC1}	β_{PC2}	β_{PC3}	β_{PC4}	β_{PC5}	β_{PC6}	β_{PC7}	β_{PC8}	$\bar{R}^2$	N
Contemporaneous ($PC_t \rightarrow R_t$)											
<i>BTP Italia</i>	1.44*** (3.30)	-0.03** (-2.25)	-0.02** (-2.02)	0.03 (1.03)	0.05 (1.45)	0.00 (0.04)	-0.02 (-0.62)	-0.04 (-1.47)	-0.02 (-1.13)	0.088	157
<i>CDS–Bond</i>	4.78*** (6.25)	0.04* (1.91)	0.06*** (2.96)	-0.09*** (-2.80)	-0.01 (-0.59)	-0.02 (-0.55)	-0.01 (-0.24)	0.04 (1.33)	0.03 (0.71)	0.109	197
<i>Index Skew</i>	2.21*** (5.95)	-0.01 (-0.83)	0.01 (0.90)	0.04*** (3.03)	0.02 (0.92)	0.04** (2.46)	-0.07*** (-2.95)	0.02 (1.07)	0.04* (1.66)	0.134	209

Table A.7. Top Factor Loadings by Principal Component

For each of the first 3 principal components, the 6 candidate factors with the largest absolute loading, with the corresponding eigenvector entry. Loadings are the full-sample eigenvectors of the standardized candidate factor panel; the sign indicates the direction of the relationship.

PC1		PC2		PC3	
Factor	Loading	Factor	Loading	Factor	Loading
CRED_SPR_EU	+0.20	GLOBAL_AGG	+0.30	SNRFIN_MAIN	+0.30
DEF_US	+0.20	GLOBAL_TERM	+0.28	BTP_BUND_2Y	+0.21
DEF_EU	+0.19	R10_EU	+0.28	PB_CDS_5Y_EU	+0.20
CRED_SPR_US	+0.19	GOVT_EU	+0.28	CREDIT_US	-0.20
ITRX_MAIN	+0.19	TERM_EU	+0.28	BTP_BUND	+0.19
HY_CORP	+0.19	RI_EU	+0.27	FI_VAL	+0.19

Table A.8. Out-of-Sample Alpha: PCA and Best-Subset

Out-of-sample alpha of the PCA (Panel A) and best-subset (Panel B) hedges, in the recursive design of Welch and Goyal (2008). At each month t the hedge is estimated on a past-only window—expanding with a 60-month burn-in (headline) or rolling 60-month, which relaxes the constant-loadings assumption—and the realized abnormal return is $e_{i,t} = r_{i,t} - \hat{\beta}'_{i,t-1} \mathbf{f}_t$; the in-window intercept (the alpha under measurement) is excluded from the hedge. In Panel A the principal components are re-estimated within each window, so the design is invariant to the sign and rotation indeterminacy of PCA; in Panel B the best-subset factor set is frozen and only the loadings are re-estimated. α is the annualized mean of $e_{i,t}$ (% p.a.), tested with Newey–West HAC standard errors; IR is the annualized information ratio $\bar{e}_i / \sigma(e_i) \times \sqrt{12}$; N is the number of out-of-sample months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Strategy	N	Expanding (60-month burn-in)			Rolling 60-month		
		α	t	IR	α	t	IR
Panel A: PCA							
<i>BTP Italia</i>	97	+1.61***	2.86	+0.97	+1.65***	3.00	+1.02
<i>CDS–Bond Basis</i>	137	+4.55***	4.98	+1.55	+3.88***	3.75	+1.25
<i>iTraxx Skew</i>	149	+1.85***	3.75	+1.11	+1.96***	3.97	+1.13
Panel B: Best-Subset							
<i>BTP Italia</i>	97	+2.05***	4.06	+1.35	+2.13***	3.95	+1.32
<i>CDS–Bond Basis</i>	137	+3.90***	4.43	+1.42	+3.74***	3.90	+1.32
<i>iTraxx Skew</i>	149	+2.50***	5.35	+1.60	+1.90***	3.52	+1.01

Table A.9. Post-Selection OLS on the Best-Subset Factors

The table reports OLS estimates of $r_{i,t} = \alpha_i + \sum_j \beta_{i,j} F_{j,t} + \varepsilon_{i,t}$, where $r_{i,t}$ is the monthly excess return of arbitrage strategy i and $F_{j,t}$ are the factors selected for that strategy by best-subset selection (Bertsimas, King and Mazumder, 2016). Factors enter in original (un-standardized) units; α is annualized (% p.a.). The 95% CI is the percentile interval for α from a stationary bootstrap (9999 replications). t -statistics use Newey–West HAC standard errors (4 lags). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Factor definitions are in Table ??.

	Coefficient	t -stat	95% CI
<i>Panel A: BTP Italia</i>			
α (% p.a.)	2.2***	5.59	[1.3, 3.0]
DEF_EU	0.058**	2.43	
LIQ_V	−0.016*	−1.88	
MKT_EU	−0.038***	−3.99	
SMB_EU	−0.056***	−2.79	
FI_VAL	−0.094**	−2.28	
FI_DEF	−0.207***	−4.94	
5Y5Y_INFL_US	−0.232***	−2.70	
SS5Y	−0.010**	−2.26	
$T = 157, k = 8, \bar{R}^2 = 0.31$			
<i>Panel B: CDS–Bond Basis</i>			
α (% p.a.)	4.8***	5.87	[2.9, 6.6]
PB_CDS_5Y_EU	−0.223**	−2.28	
DEF_US	0.112***	3.21	
HY_CORP	−0.109**	−2.20	
EMERG_DEBT	0.074**	2.17	
RI_EU	0.217***	3.06	
FX_MOM	−0.041	−1.33	
PTFSFX	−0.006**	−2.26	
CURV_2S5S10S_EU	−0.015*	−1.82	
$T = 197, k = 8, \bar{R}^2 = 0.21$			
<i>Panel C: iTraxx Skew</i>			
α (% p.a.)	2.9***	5.30	[1.5, 3.8]
CDX_IG	0.167**	2.40	
HKM_IC	−0.025***	−4.14	
HML_EU	−0.026*	−1.91	
BAB_EU	−0.044***	−4.26	
GMOM	−0.052***	−3.05	
FI_MOM	−0.041***	−3.87	
SVS_RET_1M	−0.857**	−2.52	
ATM_IV_ITRX	0.304	1.32	
$T = 209, k = 8, \bar{R}^2 = 0.25$			

Table A.10. Alpha by Stress Regime: PCA and Best-Subset

Annualized alpha (% p.a.) per stress regime, from a single regression with three intercept dummies and betas common across regimes (no constant), $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$, Newey–West HAC standard errors (4 lags). PCA: spanning regression on the $K = 8$ full-sample principal components. Best-Subset: post-selection OLS on the selected factors (Table A.9). Regimes on the 5-year iTraxx Europe Main spread: LOW < 60, MEDIUM $\in [60, 100)$, HIGH ≥ 100 bps. t -statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Strategy	PCA			Best-Subset		
	LOW	MEDIUM	HIGH	LOW	MEDIUM	HIGH
<i>BTP Italia</i>	+0.37 (0.65)	+1.68*** (3.17)	+3.27* (1.82)	+1.28** (2.29)	+2.40*** (5.08)	+3.77*** (2.95)
<i>CDS–Bond Basis</i>	+2.87*** (3.12)	+4.44*** (3.87)	+7.55*** (4.84)	+2.69*** (2.99)	+4.69*** (3.95)	+7.59*** (4.93)
<i>iTraxx Skew</i>	+1.03* (1.95)	+2.23*** (4.62)	+3.45*** (4.32)	+2.48*** (3.95)	+2.94*** (4.38)	+3.39*** (4.14)

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D Cross-Strategy Interdependencies

The complete slow-moving capital test battery: unconditional and regime-dependent correlations, interaction regressions, cross-strategy spanning regressions, Granger causality across lag lengths, AR(1) persistence, funding-stress regressions, generalized impulse responses, and DCC-GARCH dynamic correlations.

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Figure A.1. Generalized impulse response functions of the three mispricings. Shaded areas: 95% bootstrap confidence intervals.

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Figure A.2. DCC-GARCH dynamic conditional correlations of mispricing changes. Shading marks the stress regime (iTraxx Main 5Y thresholds at 60 and 100 bps).

E Data and Code Availability

Data sources: Borsa Italiana MOT (BTP Italia and nominal BTP prices), Bloomberg (inflation swaps, corporate bond spreads), CMA/CME (single-name CDS), J.P. Morgan DataQuery cross-validated against Barclays Live (iTraxx index and constituents). The full replication pipeline (Python, generating every table and figure in this report) is available from the author upon request.